

boilers 4 rows one row = one boiler · the asset register <table><tr><td> boiler_id PK internal id, 1–4 </td></tr><tr><td> code plant tag 'B-2101' </td></tr><tr><td> name 'North watertube' </td></tr><tr><td> maker_model 'Babcock FM-120' </td></tr><tr><td> rating_tph design output, t/h </td></tr><tr><td> design_barg design pressure, barg </td></tr><tr><td> commissioned year in service </td></tr><tr><td> fuel 'Natural gas' </td></tr></table>	boiler_id PK internal id, 1–4	code plant tag 'B-2101'	name 'North watertube'	maker_model 'Babcock FM-120'	rating_tph design output, t/h	design_barg design pressure, barg	commissioned year in service	fuel 'Natural gas'	operators 6 rows one row = one person on the roster <table><tr><td> operator_id PK 1 ... 6 </td></tr><tr><td> emp_no 'EM-4471' </td></tr><tr><td> full_name 'Somchai Srisuk' </td></tr><tr><td> shift 'A', 'B' or 'C' </td></tr><tr><td> cert_class 'Class 1' / 'Class 2' </td></tr></table> Values in 'quotes' are text — everything else is a number.	operator_id PK 1 ... 6	emp_no 'EM-4471'	full_name 'Somchai Srisuk'	shift 'A', 'B' or 'C'	cert_class 'Class 1' / 'Class 2'	water_tests 45 rows one row = one weekly water sample <table><tr><td> test_id PK 1 ... 45 </td></tr><tr><td> boiler_id FK → boilers.boiler_id </td></tr><tr><td> test_date 'YYYY-MM-DD' </td></tr><tr><td> silica_ppm ppm (plant limit 10) </td></tr><tr><td> conductivity_us µS/cm </td></tr><tr><td> phosphate_ppm residual, ppm </td></tr><tr><td> ph pH of the sample </td></tr></table>	test_id PK 1 ... 45	boiler_id FK → boilers.boiler_id	test_date 'YYYY-MM-DD'	silica_ppm ppm (plant limit 10)	conductivity_us µS/cm	phosphate_ppm residual, ppm	ph pH of the sample
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readings 1,866 rows one row = one instrument reading, every 4 h <table><tr><td> reading_id PK 1 ... 1866 </td></tr><tr><td> boiler_id FK → boilers.boiler_id </td></tr><tr><td> ts 'YYYY-MM-DD HH:MM' </td></tr><tr><td> steam_tph steam flow, t/h </td></tr><tr><td> stack_temp_c flue gas, °C </td></tr><tr><td> o2_pct flue gas O₂, % </td></tr></table>	reading_id PK 1 ... 1866	boiler_id FK → boilers.boiler_id	ts 'YYYY-MM-DD HH:MM'	steam_tph steam flow, t/h	stack_temp_c flue gas, °C	o2_pct flue gas O₂, %	daily_logs 311 rows one row = one boiler on one day <table><tr><td> log_id PK 1 ... 311 </td></tr><tr><td> boiler_id FK → boilers.boiler_id </td></tr><tr><td> operator_id FK → operators.operator_id </td></tr><tr><td> log_date 'YYYY-MM-DD' </td></tr><tr><td> steam_t steam that day, t </td></tr><tr><td> fuel_gj gas that day, GJ </td></tr></table>	log_id PK 1 ... 311	boiler_id FK → boilers.boiler_id	operator_id FK → operators.operator_id	log_date 'YYYY-MM-DD'	steam_t steam that day, t	fuel_gj gas that day, GJ	maintenance_events 14 rows one row = one job done on one boiler <table><tr><td> event_id PK 1 ... 14 </td></tr><tr><td> boiler_id FK → boilers.boiler_id </td></tr><tr><td> event_date 'YYYY-MM-DD' </td></tr><tr><td> event_type 'Tube cleaning', 'Burner service', 'Inspection', ... </td></tr><tr><td> hours_down time off line, h </td></tr><tr><td> notes what the fitter wrote </td></tr></table>	event_id PK 1 ... 14	boiler_id FK → boilers.boiler_id	event_date 'YYYY-MM-DD'	event_type 'Tube cleaning', 'Burner service', 'Inspection', ...	hours_down time off line, h	notes what the fitter wrote		
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fuel_prices 3 rows one row = what gas cost in one month <table><tr><td> month PK '2026-02' ... '2026-04' </td></tr><tr><td> baht_per_gj gas price, ฿/GJ </td></tr></table> Join it on a month you cut out yourself: ON f.month = substr(d.log_date,1,7)	month PK '2026-02' ... '2026-04'	baht_per_gj gas price, ฿/GJ	boiler_id is not the tag read this twice 1 = B-2101 North · 2 = B-2103 South 3 = B-2102 Centre · 4 = B-2104 New The big tables store only the id, so never name a machine from its number — JOIN boilers b ON b.boiler_id = r.boiler_id and read the tag off b.code.	clause order always this order WITH a named sub-query optional SELECT which columns you want FROM which table JOIN...ON how another table matches WHERE which rows to keep GROUP BY what to summarise by HAVING which groups to keep ORDER BY how to sort DESC = big first LIMIT how many rows to show Only SELECT and FROM are compulsory.																		
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From the question to the query			
“How many ...?”	→ COUNT(*) · COUNT(DISTINCT x)	“The worst / the top five”	→ ORDER BY x DESC + LIMIT 5
“For each boiler / shift / day”	→ GROUP BY that column	“Show the tag, not the id”	→ JOIN the small table
“Only where ...”	→ WHERE rows · HAVING groups	“How often is it above X?”	→ WHERE x > X then COUNT(*)
“Per tonne / per hour / per day”	→ divide one SUM by another	“Compared with the average”	→ > (SELECT AVG(x) FROM t) in the WHERE
“Why did it change?”	→ look it up in maintenance_events	· next step	
		“Is it getting worse?”	→ GROUP BY substr(ts, 1, 7) — by month
		· next step	

Traps that catch everybody - Text needs single quotes: code = 'B-2103', never "B-2103". Numbers take none. - WHERE filters rows, HAVING filters groups — WHERE AVG(x)>150 is an error. - Check MIN and MAX before you trust an AVG — a dead thermocouple reading 0.0 and a spike of 511.8 are both hiding in readings. - A rate is SUM(fuel_gj)/SUM(steam_t), not AVG(fuel_gj/steam_t) — the second lets a quiet day count as much as a busy one. - Anything in SELECT beside an aggregate must appear in GROUP BY too. - Put COUNT(*) next to every average — unequal counts mean an outage, a new machine, or missing data. - A join need not be on an id — ON f.month = substr(log_date,1,7) works.	The shape of a real query <pre>-- which boiler wastes gas? one row per boiler SELECT b.code, ROUND(SUM(d.fuel_gj)/SUM(d.steam_t), 3) AS gj_t, COUNT(*) AS n_days FROM daily_logs d JOIN boilers b ON b.boiler_id = d.boiler_id WHERE d.log_date LIKE '2026-04%' GROUP BY b.code HAVING COUNT(*) > 20 ORDER BY gj_t DESC Pieces worth remembering: AVG MIN MAX SUM COUNT · ROUND(x, 1) · substr(ts, 1, 10) the day · substr(ts, 1, 7) the month · x IN ('B','C') · (SELECT AVG(x) FROM t) as one value</pre>
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